

Coffee Shop Sales Analysis

Project Documentation

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1. Introduction

This project analyses transactional sales data from a coffee shop maven roaster at NYC. The objective is to uncover sales trends, evaluate product performance, and provide actionable business recommendations using Excel-based analysis.

2. Aim

To identify key sales drivers, optimize pricing strategies, and support data-driven decision-making for the coffee shop.

3. Business Problem / Context

The coffee shop wants to maximize revenue and customer satisfaction by understanding which products sell best, when sales peak, and how pricing changes impact overall sales.

4. Project Workflow

- The data is taken from the Kaggle dataset and imported to Excel.
- Explored and understood the dataset.
- Removed blank cells and formatted columns.
- Performed calculations on the dataset for analysis.
- Filter and segment data (e.g., store location, timeline).
- Perform statistical and formula-based analysis.
- Pivot tables and charts.
- Apply scenario analysis (What-If, Goal Seek, Scenario Manager).
- Summarize insights and provide recommendations.

5. Data Understanding

- The Coffee Shop sales Dataset size is 1L+ rows and 11 columns.
- It has Columns like Transaction Id, Transaction Date, Transaction Time, Unit Price, Quantity, Product Item etc., and its types are as Date, Text, Number, Quantity (Number)

transaction_id	transaction_date	transaction_time	transaction_qty	store_id	store_location	product_id	unit_price	product_category	product_type	product_detail
1	01-01-2023	07:06:11	2	5	Lower Manhattan	32	3	Coffee	Gourmet brewed coffee	Ethiopia Rg
2	01-01-2023	07:08:56	2	5	Lower Manhattan	57	3.1	Tea	Brewed Chai tea	Spicy Eye Opener Chai Lg
3	01-01-2023	07:14:04	2	5	Lower Manhattan	59	4.5	Drinking Chocolate	Hot chocolate	Dark chocolate Lg
4	01-01-2023	07:20:24	1	5	Lower Manhattan	22	2	Coffee	Drip coffee	Our Old Time Diner Blend Sm
5	01-01-2023	07:22:41	2	5	Lower Manhattan	57	3.1	Tea	Brewed Chai tea	Spicy Eye Opener Chai Lg
6	01-01-2023	07:22:41	1	5	Lower Manhattan	77	3	Bakery	Scone	Oatmeal Scone
7	01-01-2023	07:25:49	1	5	Lower Manhattan	22	2	Coffee	Drip coffee	Our Old Time Diner Blend Sm
8	01-01-2023	07:33:34	2	5	Lower Manhattan	28	2	Coffee	Gourmet brewed coffee	Columbian Medium Roast Sm
9	01-01-2023	07:39:13	1	5	Lower Manhattan	39	4.25	Coffee	Barista Espresso	Latte Rg
10	01-01-2023	07:39:34	2	5	Lower Manhattan	58	3.5	Drinking Chocolate	Hot chocolate	Dark chocolate Rg
11	01-01-2023	07:43:05	1	5	Lower Manhattan	56	2.55	Tea	Brewed Chai tea	Spicy Eye Opener Chai Rg
12	01-01-2023	07:44:35	2	5	Lower Manhattan	33	3.5	Coffee	Gourmet brewed coffee	Ethiopia Lg
13	01-01-2023	07:45:51	1	5	Lower Manhattan	51	3	Tea	Brewed Black tea	Earl Grey Lg
14	01-01-2023	07:48:19	1	5	Lower Manhattan	57	3.1	Tea	Brewed Chai tea	Spicy Eye Opener Chai Lg
15	01-01-2023	07:52:36	2	5	Lower Manhattan	87	3	Coffee	Barista Espresso	Ouro Brasileiro shot
16	01-01-2023	07:59:58	2	5	Lower Manhattan	47	3	Tea	Brewed Green tea	Serenity Green Tea Lg
17	01-01-2023	07:59:58	1	5	Lower Manhattan	79	3.75	Bakery	Scone	Jumbo Savory Scone
18	01-01-2023	08:00:18	1	8	Hell's Kitchen	42	2.5	Tea	Brewed herbal tea	Lemon Grass Rg
19	01-01-2023	08:00:39	2	8	Hell's Kitchen	59	4.5	Drinking Chocolate	Hot chocolate	Dark chocolate Lg
20	01-01-2023	08:11:45	1	8	Hell's Kitchen	61	4.75	Drinking Chocolate	Hot chocolate	Sustainably Grown Organic Lg

6. Data Cleaning

- Removed blank records in the dataset by checking blank rows in Go to Special option in Find & Select.

Transaction_id	Transaction_date	Transaction_time	Store_id	Store_location	Product_id	Unit_price	Product_category	Product_type	Product_detail	Total_amount	Hour_of_day	Day_of_week
1	01 January 2023	07:06 AM	2	5 Lower Ma	32	\$3.00	Coffee	Gourmet Brewed Coffee	Ethiopia Rg	\$6.00		7
2	01 January 2023	07:08 AM	2	5 Lower Ma	57	\$3.10	Tea	Brewed Chai Tea	Hot Chocolate			7
3	01 January 2023	07:14 AM	2	5 Lower Ma	59	\$4.50	Drinking Chocolate	Brewed Chai Tea				7
4	01 January 2023	07:20 AM	1	5 Lower Ma	22	\$2.00	Coffee	Drip Coffee				7
5	01 January 2023	07:22 AM	2	5 Lower Ma	57	\$3.10	Tea	Brewed Chai Tea				7
6	01 January 2023	07:22 AM	1	5 Lower Ma	77	\$3.00	Bakery	Scone				7
7	01 January 2023	07:25 AM	1	5 Lower Ma	22	\$2.00	Coffee	Drip Coffee				7
8	01 January 2023	07:33 AM	2	5 Lower Ma	28	\$2.00	Coffee	Gourmet Brewed Coffee				7
9	01 January 2023	07:39 AM	1	5 Lower Ma	39	\$4.25	Coffee	Barista Espresso				7
10	01 January 2023	07:39 AM	2	5 Lower Ma	58	\$3.50	Drinking Chocolate	Hot Chocolate				7
11	01 January 2023	07:43 AM	1	5 Lower Ma	56	\$2.55	Tea	Brewed Chai Tea				7
12	01 January 2023	07:44 AM	2	5 Lower Ma	33	\$3.50	Coffee	Gourmet Brewed Coffee				7
13	01 January 2023	07:45 AM	1	5 Lower Ma	51	\$3.00	Tea	Brewed Black Tea				7
14	01 January 2023	07:48 AM	1	5 Lower Ma	57	\$3.10	Tea	Brewed Chai Tea				7
15	01 January 2023	07:52 AM	2	5 Lower Ma	87	\$3.00	Coffee	Barista Espresso				7
16	01 January 2023	07:59 AM	2	5 Lower Ma	47	\$3.00	Tea	Brewed Green Tea				7
17	01 January 2023	07:59 AM	1	5 Lower Ma	79	\$3.75	Bakery	Scone				7
18	01 January 2023	08:00 AM	1	8 Hell's Kitc	42	\$2.50	Tea	Brewed Herbal Tea	Lemon Grass Rg	\$2.50		8
19	01 January 2023	08:00 AM	2	8 Hell's Kitc	59	\$4.50	Drinking Chocolate	Hot Chocolate	Dark Chocolate Lg	\$9.00		8
20	01 January 2023	08:11 AM	1	8 Hell's Kitc	61	\$4.75	Drinking Chocolate	Hot Chocolate	Sustainably Grown Organic Lg	\$4.75		8

- Checked for and handled missing values (none found).
- Standardized product names by the Proper () function, and Formatted Date and time field and Trim () function are used in Text Functions like Store Location, Product item , category etc.,

	A	B	C	D	E	F	G	H	I	J	K	
1	Transaction_id	Transaction_date	Transaction_time	Transaction_qty	Store_id	Store_location	Product_id	Unit_price	Product_category	Product_type	Product_detail	
2	1	01 January 2023	07:06 AM	2	5	Lower Manhattan	32	\$3.00	Coffee	Gourmet Brewed Coffee	Ethiopia Rg	
3	2	01 January 2023	07:08 AM	2	5	Lower Manhattan	57	\$3.10	Tea	Brewed Chai Tea	Spicy Eye Opener Chai Lg	
4	3	01 January 2023	07:14 AM	2	5	Lower Manhattan	59	\$4.50	Drinking Chocolate	Hot Chocolate	Dark Chocolate Lg	
5	4	01 January 2023	07:20 AM	1	5	Lower Manhattan	22	\$2.00	Coffee	Drip Coffee	Our Old Time Diner Blend Sm	
6	5	01 January 2023	07:22 AM	2	5	Lower Manhattan	57	\$3.10	Tea	Brewed Chai Tea	Spicy Eye Opener Chai Lg	
7	6	01 January 2023	07:22 AM	1	5	Lower Manhattan	77	\$3.00	Bakery	Scone	Oatmeal Scone	
8	7	01 January 2023	07:25 AM	1	5	Lower Manhattan	22	\$2.00	Coffee	Drip Coffee	Our Old Time Diner Blend Sm	
9	8	01 January 2023	07:33 AM	2	5	Lower Manhattan	28	\$2.00	Coffee	Gourmet Brewed Coffee	Columbian Medium Roast Sm	
10	9	01 January 2023	07:39 AM	1	5	Lower Manhattan	39	\$4.25	Coffee	Barista Espresso	Latte Rg	
11	10	01 January 2023	07:39 AM	2	5	Lower Manhattan	58	\$3.50	Drinking Chocolate	Hot Chocolate	Dark Chocolate Rg	
12	11	01 January 2023	07:43 AM	1	5	Lower Manhattan	56	\$2.55	Tea	Brewed Chai Tea	Spicy Eye Opener Chai Rg	
13	12	01 January 2023	07:44 AM	2	5	Lower Manhattan	33	\$3.50	Coffee	Gourmet Brewed Coffee	Ethiopia Lg	
14	13	01 January 2023	07:45 AM	1	5	Lower Manhattan	51	\$3.00	Tea	Brewed Black Tea	Earl Grey Lg	
15	14	01 January 2023	07:48 AM	1	5	Lower Manhattan	57	\$3.10	Tea	Brewed Chai Tea	Spicy Eye Opener Chai Lg	
16	15	01 January 2023	07:52 AM	2	5	Lower Manhattan	87	\$3.00	Coffee	Barista Espresso	Ouro Brasileiro Shot	
17	16	01 January 2023	07:59 AM	2	5	Lower Manhattan	47	\$3.00	Tea	Brewed Green Tea	Serenity Green Tea Lg	
18	17	01 January 2023	07:59 AM	1	5	Lower Manhattan	79	\$3.75	Bakery	Scone	Jumbo Savory Scone	
19	18	01 January 2023	08:00 AM	1	8	Hell'S Kitchen	42	\$2.50	Tea	Brewed Herbal Tea	Lemon Grass Rg	
20	19	01 January 2023	08:00 AM	2	8	Hell'S Kitchen	59	\$4.50	Drinking Chocolate	Hot Chocolate	Dark Chocolate Lg	
21	20	01 January 2023	08:11 AM	1	8	Hell'S Kitchen	61	\$4.75	Drinking Chocolate	Hot Chocolate	Sustainably Grown Organic Lg	

• 7. Derived Calculations & Metrics

- To perform the analysis, derived some calculations and metrics on the dataset by using the Aggregate, Text, Date and Time functions.
- Total Amount: =Price * Quantity
- Quarter: =ROUNDUP(MONTH(Date)/3,0)
- Hour, Day, Weekday and Number by Hour (), Weekday (), Week Number (), Text ()
- Top Revenue and Item sold is calculated by sum ()
- Peak hours, day and Top product are calculated by index (), unique ()

	L	M	N	O	P	Q	R	S	T	U	V	W
	Total_amount	Hour_of_day	Day_of_week	Day_name	Month_nam	Quarter	Week_number					
	\$6.00	7	1	Sunday	January	1	1					
	\$6.20	7	1	Sunday	January	1	1		Total_revenue			
	\$9.00	7	1	Sunday	January	1	1		\$6,98,812.33			
	\$2.00	7	1	Sunday	January	1	1					
	\$6.20	7	1	Sunday	January	1	1		Items_sold			
	\$3.00	7	1	Sunday	January	1	1		214470			
	\$2.00	7	1	Sunday	January	1	1					
	\$4.00	7	1	Sunday	January	1	1		Peak_hours			
	\$4.25	7	1	Sunday	January	1	1		10			
	\$7.00	7	1	Sunday	January	1	1					
	\$2.55	7	1	Sunday	January	1	1		Peak_day			
	\$7.00	7	1	Sunday	January	1	1		Monday			
	\$3.10	7	1	Sunday	January	1	1		Top_product			
	\$6.00	7	1	Sunday	January	1	1		Sustainably Grown Organic Lg			
	\$6.00	7	1	Sunday	January	1	1					
	\$3.75	7	1	Sunday	January	1	1					
	\$2.50	8	1	Sunday	January	1	1					
	\$9.00	8	1	Sunday	January	1	1					
	\$4.75	8	1	Sunday	January	1	1					

8. Filtering / Segmentation

- Applied filters to analyze sales data by store location and time periods (monthly, quarterly, daily) to identify performance patterns across different geographies and timeframes.



- Segmented sales data by product categories and individual items to identify best-selling products based on revenue contribution and sales volume. This multi-dimensional analysis enabled data-driven insights for inventory management, resource allocation, and strategic decision-making across all store locations.



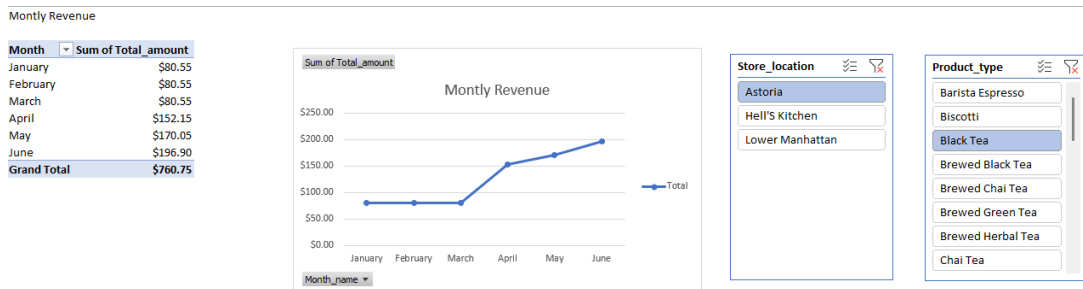
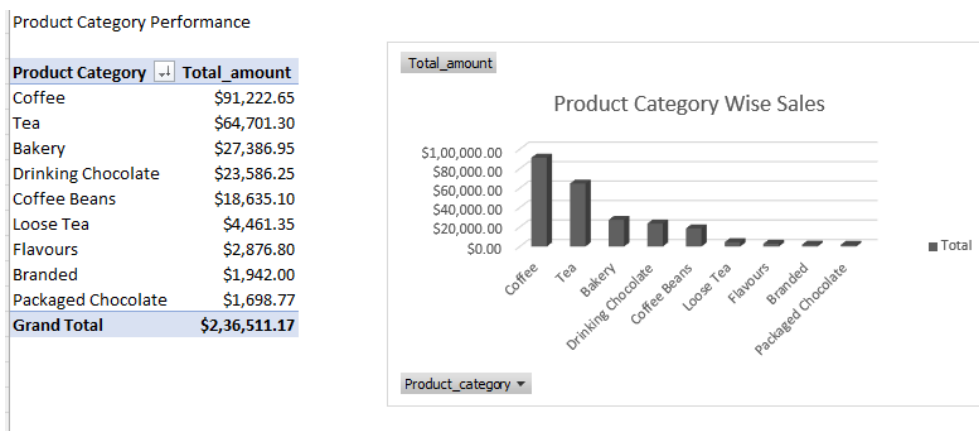
9. Statistical / Formula-Based Analysis

- Calculated total, sales, revenue, top product and peak hour and day by using Functions.
- Used SUMIFS, INDEX-MATCH, and UNIQUE, SUM formulas for advanced metrics.

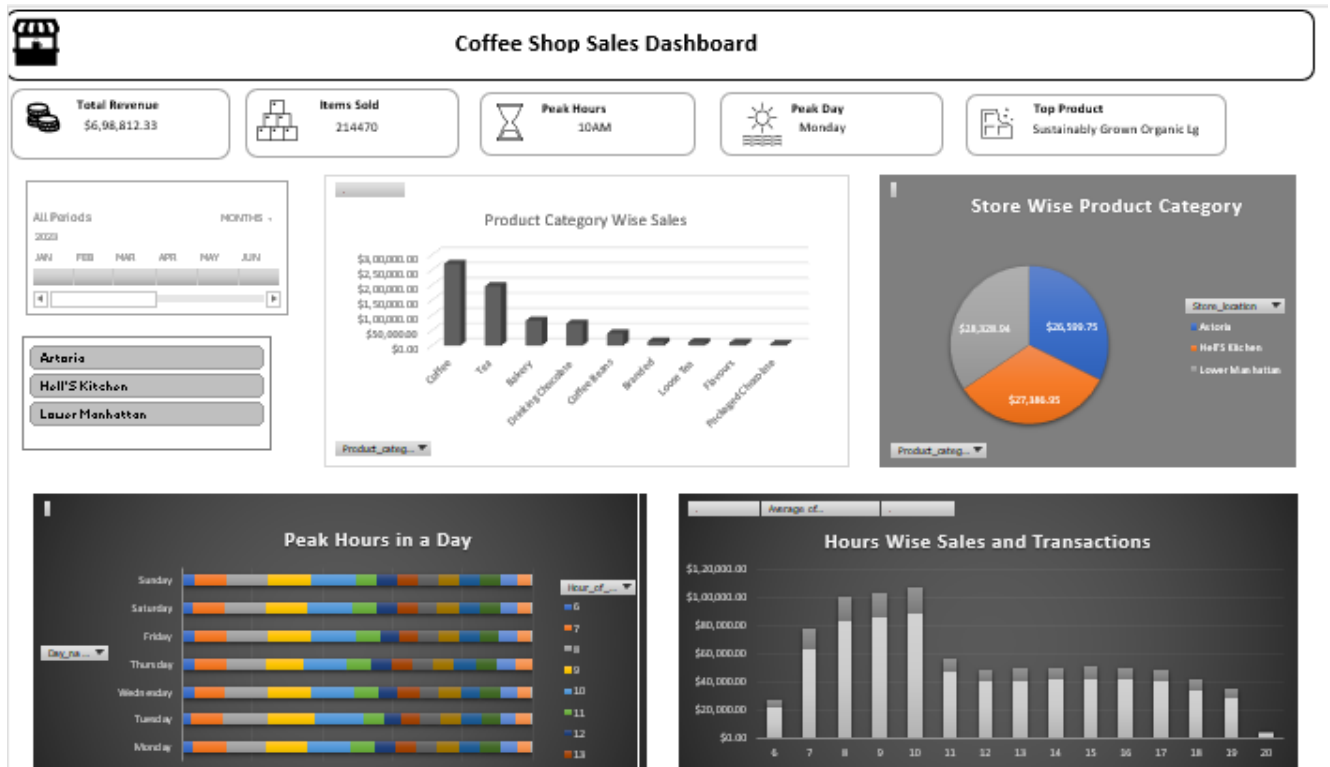
```
=INDEX(UNIQUE(Calculations!M:M),MATCH(MAX(SUMIFS(Calculations!L:L,Calculations!M:M,UNIQUE(Calculations!M:M))),SUMIFS(Calculations!L:L,Calculations!M:M,UNIQUE(Calculations!M:M)),0))
```

10. PivotTables, Pivot Charts, Visualizations

- PivotTables for dynamic sales analysis by product, month, and location, and created correspondingly.
- Pivot Charts to visualize sales trends, identify top products, and compare monthly performance through interactive dashboards.
- Example: Product Category and Monthly revenue pivot tables are shown and it's corresponding pivot charts in bar and line charts.



- Dashboard:
Built a dynamic dashboard with KPIs, charts, and slicers to visualize total sales, product performance, store comparisons, and temporal trends, providing stakeholders with an at-a-glance view of business performance across all dimensions.



11. Scenario Analysis

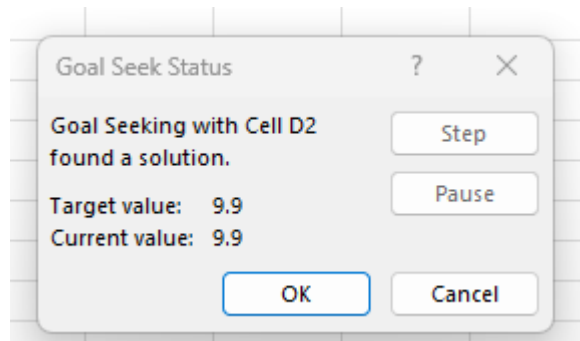
- What-If-Analyse:

To evaluate the impact of price changes on total sales revenue by creating data tables that model various pricing scenarios for the item Coffee.

Product It	Quantity	Price	Sales
Coffee	2	3.1	6.2
Price	Total Sales		
2	6.2		
4.5	9		
6.8	13.6		
9.1	18.2		
5.2	10.4		

- Goal Seek:

Utilized Excel's Goal Seek function to perform reverse calculation, determining the exact product price required to achieve a specific target sales revenue. By setting the desired sales goal, the tool automatically calculated the necessary price adjustment, providing precise pricing recommendations to meet business objectives.



- **Scenario Manager:**
Created and compared multiple pricing strategy scenarios (Likely High Sales, Low Sales) using Scenario Manager to evaluate different combinations of variables such as product prices, Quantity, and Sales. Generated a comprehensive scenario summary report that compares outcomes side-by-side, facilitating strategic decision-making by visualizing potential financial impacts under different market conditions.

Scenario Summary			
	Current Values:	High Sales	Low Sales
Changing Cells:			
\$B\$2	2.1	9.1	2.1
Result Cells:			
\$D\$2	6.3	27.3	6.3

Notes: Current Values column represents values of changing cells at time Scenario Summary Report was created. Changing cells for each scenario are highlighted in gray.

12. Power Pivot & Data Modelling

This Power Pivot model creates a relational database structure linking Transaction and Product tables through diagram view relationships. DAX measures were implemented to calculate key metrics like total revenue and sales performance. The resulting Power Pivot table enables dynamic, multi-dimensional analysis of coffee shop sales data with enhanced performance compared to standard Excel pivot tables.

Power Pivot for Excel - Coffee Shop Sales - Copy.xlsx

File Home Design Advanced

Paste Get External Data Refresh PivotTable Data Type: Format: General \$ % > < .00 .00 Clear All Filters Sort by Column Find AutoSum Create KPI Diagram View Show Hidden Calculation Area

Clipboard f_x Measure 1:=COUNT((Transaction_qty))

	Transaction_id	Transaction_date	Transaction_time	Transaction_qty	Product_id	Add Column
1	1	01-01-2023 00:00:00	30-12-1899 07:06:11	2	32	
2	2	01-01-2023 00:00:00	30-12-1899 07:08:56	2	57	
3	3	01-01-2023 00:00:00	30-12-1899 07:14:04	2	59	
4	4	01-01-2023 00:00:00	30-12-1899 07:20:24	1	22	
5	5	01-01-2023 00:00:00	30-12-1899 07:22:41	2	57	
6	6	01-01-2023 00:00:00	30-12-1899 07:22:41	1	77	
7	7	01-01-2023 00:00:00	30-12-1899 07:25:49	1	22	
8	8	01-01-2023 00:00:00	30-12-1899 07:33:34	2	28	
9	9	01-01-2023 00:00:00	30-12-1899 07:39:13	1	39	
10	10	01-01-2023 00:00:00	30-12-1899 07:39:34	2	58	
11	11	01-01-2023 00:00:00	30-12-1899 07:43:05	1	56	
12	12	01-01-2023 00:00:00	30-12-1899 07:44:35	2	33	
13	13	01-01-2023 00:00:00	30-12-1899 07:45:51	1	51	
14	14	01-01-2023 00:00:00	30-12-1899 07:48:19	1	57	
15	15	01-01-2023 00:00:00	30-12-1899 07:52:36	2	87	
16	16	01-01-2023 00:00:00	30-12-1899 07:59:58	2	47	
17	17	01-01-2023 00:00:00	30-12-1899 07:59:58	1	79	

Measure 1: 30

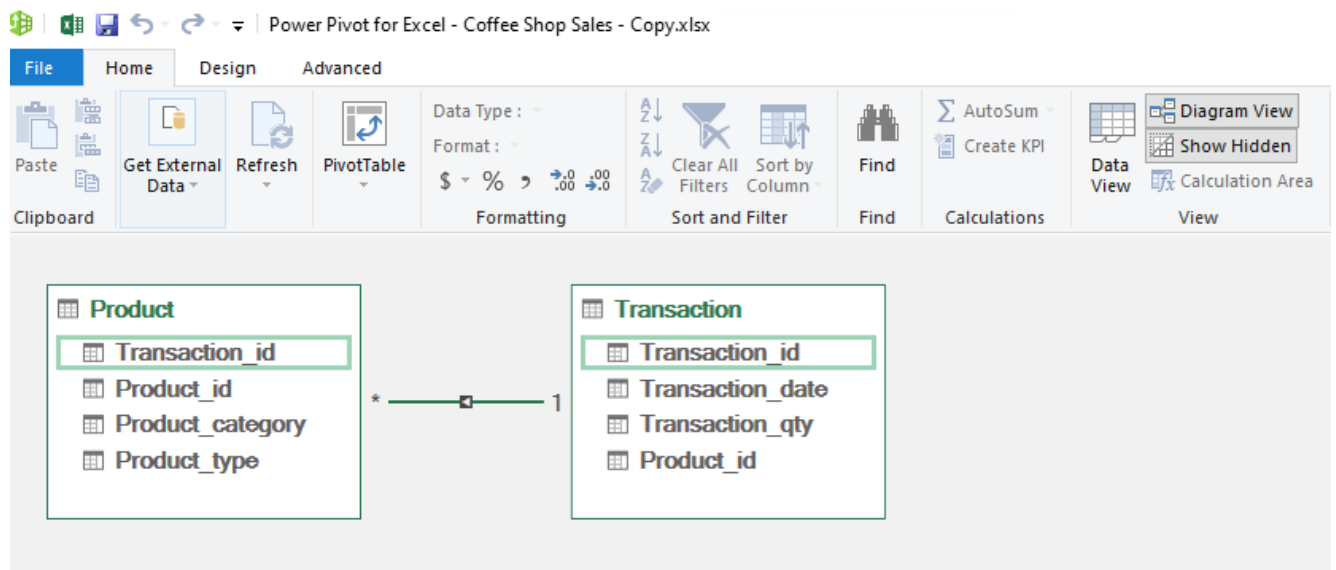
Power Pivot for Excel - Coffee Shop Sales - Copy.xlsx

File Home Design Advanced

Paste Get External Data Refresh PivotTable Data Type: Format: General \$ % > < .00 .00 Clear All Filters Sort by Column Find AutoSum Create KPI Diagram View Show Hidden Calculation Area

Clipboard f_x

	Transaction_id	Product_id	Product_category	Product_type	Add Column
1	1	32	Coffee	Gourmet brew...	
2	2	57	Tea	Brewed Chai tea	
3	3	59	Drinking Chocolate	Hot chocolate	
4	4	22	Coffee	Drip coffee	
5	5	57	Tea	Brewed Chai tea	
6	6	77	Bakery	Scone	
7	7	22	Coffee	Drip coffee	
8	8	28	Coffee	Gourmet brew...	
9	9	39	Coffee	Barista Espres...	
10	10	58	Drinking Chocolate	Hot chocolate	
11	11	56	Tea	Brewed Chai tea	
12	12	33	Coffee	Gourmet brew...	
13	13	51	Tea	Brewed Black t...	
14	14	57	Tea	Brewed Chai tea	
15	15	87	Coffee	Barista Espres...	
16	16	47	Tea	Brewed Green ...	



13. Insights

- **Peak Sales Periods:**
Sales are highest during the morning hours and at weekends, indicating strong demand during breakfast and leisure times.
- **Top-Selling Products:**
Espresso/Coffee-based drinks consistently generate the most revenue, while specialty drinks have lower but steady sales.

14. Recommendations

- **Promote Best Sellers:**
Highlight top-selling products with in-store promotions and loyalty rewards to maximize revenue.
- **Bundle Offers:**
Create more combo deals to encourage higher average transaction values.

15. Conclusion & Recommendations

Strategic focus should be placed on promoting top-performing products (espresso drinks) during peak periods (mornings and weekends), while leveraging seasonal marketing campaigns to capitalize on colder months' increased demand. Expanding combo deal offerings and maintaining regular dashboard updates will ensure data-driven decision-making for sustained business growth.